PREHOSPITAL STROKE PROTOCOL

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Comprehensive Stroke Center (CSC) Referral Network

Aligned with the 2026 AHA/ASA Guideline for the Early Management of Patients with Acute Ischemic Stroke

1. Dispatch & Public Awareness

- **[1]** Sustained public education programs on stroke recognition (FAST/BE-FAST) and the need to call 997 (Saudi Red Crescent Authority), targeted to diverse communities and reinforced over time.
- **[2a]** SRCA dispatchers should use a validated telephone stroke assessment tool to support early identification and prioritize dispatch.
- **[1]** Targeted stroke education for SRCA professionals and physicians to reduce prehospital delay and maximize treatment eligibility.

2. On-Scene Assessment

2.1 Stroke Identification

- **[1]** Use a brief, validated stroke screening tool (e.g., Cincinnati Prehospital Stroke Scale) for all patients with suspected stroke.
- **[1]** Apply a validated LVO severity scale (RACE, VAN, LAMS, or FAST-ED) to identify candidates for direct thrombectomy-center transport.

2.2 Core Data to Capture

- **Last known well (LKW) time** — establish precisely; drives the entire treatment window
- **Point-of-care blood glucose** (rule out hypo/hyperglycemia mimics)
- **Baseline vital signs**, including SpO_2
- **Focused history**: anticoagulant/antiplatelet use, recent surgery or trauma, seizure at onset
- **Witness/bystander contact information** for onset time verification

3. Field Interventions

3.1 Recommended

- **[1]** Airway support and ventilatory assistance as needed for decreased consciousness or bulbar dysfunction.
- **[1]** Supplemental oxygen only if hypoxic, titrated to $SpO_2 > 94\%$.
- **[1]** Correct hypotension/hypovolemia to maintain systemic perfusion.

3.2 Do Not Perform (Class 3 — No Benefit / Harm)

- **AVOID —** Remote ischemic conditioning via arm BP cuff inflation — no functional benefit.
- **AVOID —** Prehospital transdermal glyceryl trinitrate (nitroglycerin) — potentially harmful.
- **AVOID —** Intensive field BP lowering to 130–140 mmHg systolic — no functional benefit.
- **AVOID —** Routine supplemental oxygen in non-hypoxic patients — no benefit.

4. Advance Notification

- [1] Provide advance notification to the receiving hospital for every suspected stroke transport — reduces in-hospital evaluation time, increases thrombolytic use, and decreases mortality.

Notification should include: LKW time, stroke scale/LVO severity score, vitals, glucose, anticoagulant status, and estimated time of arrival, communicated per regional stroke alert protocol.

5. SRCA Destination Decision-Making

- [1] Transport to the closest appropriate stroke-capable facility (ASRH, PSC, TSC, or CSC) rather than a non-stroke-capable hospital.

5.1 Areas with Local Access to a Thrombectomy-Capable Center

- [2a] For suspected LVO stroke, direct transport to a thrombectomy-capable center (TSC/CSC) over initial transport to a non-TSC with secondary transfer.

5.2 Areas with a Well-Coordinated Stroke System of Care

- **AVOID —** Bypassing a local, thrombolysis-capable stroke center for a distant (45–60 min) thrombectomy center does not improve 3-month outcomes where transfer networks are reliable — "drip-and-ship" remains appropriate.

5.3 Areas Without a Well-Coordinated Stroke System

- [2b] Direct transport of suspected LVO patients to the closest thrombectomy-capable center may be reasonable, provided it will not disqualify the patient from IV thrombolysis.

5.4 Interhospital Transfer

- [1] pre-established transfer agreements and protocols between spoke facilities and the CSC to minimize door-in-door-out (DIDO) time.

6. Mobile Stroke Units (If Deployed in Network)

- [1] Use MSUs over conventional SRCA transport, where available, for thrombolytic-eligible patients to minimize onset-to-treatment time.
- [1] MSUs must be equipped to diagnose and administer IV thrombolysis in the field.
- [1] MSU care should include streamlined protocols and neurological expertise (in-person or telemedicine).
- [2a] For EVT-eligible patients, MSUs can help identify and triage to the correct thrombectomy-capable facility with prehospital notification of the stroke team.

7. Prehospital Telemedicine

- [1] Consider in-ambulance telemedicine, when feasible, to complement paramedic assessment and identify candidates for reperfusion therapy.

8. CSC Arrival & Handoff

- **Stroke team activated** in advance based on prehospital notification (physician, nursing, imaging, lab)
- **NIHSS performed immediately** on arrival
- **Emergent non-contrast CT ± CTA** — target completion within 25 minutes of arrival
- **Confirm point-of-care glucose**
- **Consider direct-to-angiography-suite pathway** for LVO patients meeting severity criteria (e.g., RACE > 4) and EVT eligibility
- **Telestroke/teleradiology support** available 24/7 for imaging review and treatment decision-making at all network sites

9. Time-to-Treatment Targets

Metric	Target
Door-to-CT (imaging) completed	≤ 25 minutes
Door-to-needle (IVT)	≤ 60 minutes (≤ 45 min goal for CSC)
Door-to-puncture (EVT groin/access)	As fast as achievable — tracked per case
Door-in-door-out (transferring facility)	≤ 60 minutes
SRCA on-scene time (suspected stroke)	≤ 15 minutes when feasible

10. Quality Improvement & Metrics

- [1] Track door-to-puncture time, successful reperfusion (mTICI ≥2b), and long-term patient outcomes for all EVT cases.
- [1] Credential neurointerventionalists against established, agreed-upon training and certification standards.
- [1] Participate in a stroke data registry and engage in continuous, multicomponent QI review to reduce disparities and improve outcomes.

Recommendation Class Legend

[1] Strong (benefit >>> risk) [2a] Moderate (benefit >> risk) [2b] Weak (benefit ≥ risk)
[3H] Harm — should not be performed [3N] No benefit — not recommended

Source: Prabhakaran S, et al. 2026 Guideline for the Early Management of Patients With Acute Ischemic Stroke: A Guideline From the American Heart Association/American Stroke Association. Stroke. 2026;10.1161/STR.0000000000000513. Replaces the 2018 AIS Guideline and 2019 update.

SRCA & AHA/ASA Regulatory Alignment and Accreditation

Guideline concordance: This protocol's recommendation classes and field interventions are drawn directly from the 2026 AHA/ASA Guideline for the Early Management of Patients with Acute Ischemic Stroke, with no deviations from Class 1/2a/2b guidance or Class 3 harm/no-benefit designations.

SRCA regulatory alignment: Dispatch triage, on-scene time targets, destination decision-making, and advance notification requirements in Sections 1, 2, 4, and 5 are aligned with Saudi Red Crescent Authority (SRCA) national ambulance response and prehospital stroke transport standards, ensuring dual compliance with national regulatory requirements and international clinical guidelines.

Accreditation support: Advance notification, LVO screening, DIDO, and door-to-treatment metrics in Sections 2, 4, 5, and 9 are structured to meet Comprehensive Stroke Center (CSC) certification requirements (e.g., American Stroke Association International Stroke Center Certification Advanced Certification, AHA Mission: Lifeline® Stroke, and Get With The Guidelines®-Stroke reporting elements).

Reference standard: Prabhakaran S, et al. 2026 AHA/ASA Guideline for the Early Management of Patients With Acute Ischemic Stroke (see source citation below); Saudi Red Crescent Authority (SRCA) National Ambulance and Prehospital Stroke Response Standards.

Approval & Sign-Off

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